

Program : Diploma in Communication and Computer Networking	
Course Code : 6322B	Course Title: Fundamentals of Python
Semester : 6	Credits: 4
Course Category:	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- Familiarize python environment and basic building blocks of python
- Illustrates uses of conditional and iteration statements in python programs and
- understand the working of Strings in Python
- Develop programs utilizing modules lists, dictionaries, sets,tuples.
- Illustrate the use of numpy module and how to handle exceptions in python

Course Prerequisites: NIL

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize python environment and basic building blocks of python	14	Understanding
CO2	Illustrates uses of conditional and iteration statements in python programs and understand the working of Strings in Python	14	Understanding
CO3	Develop programs utilizing modules lists, dictionaries, sets, tuples.	16	Applying
CO4	Illustrate the use of numpy module in python	14	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	2	2				
CO2	2	3	2				
CO3	2	3	3	2			
CO4	2	2	2	2			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize python environment and basic building blocks of python		
M1.01	To understand how to install Python IDE and editor.	2	Understanding
M1.02	To understand how to run simple python script	2	Understanding
M1.03	To understand Variables, Keywords and constants in Python	4	Understanding
M1.04	To understand how to use indentation in python	1	Understanding
M1.05	To understand data types and different operators used in python	5	Understanding
Contents: Introduction to Python: Getting started with Python-Python environment setup - Installation and working of IDLE, Detecting and correcting syntax errors, How Python works, Features of Python - Interactive, Object-oriented, interpreted, platform independent. Applications of python Python building blocks - Identifiers, Keywords, Indentation, Variables, Data Types, Operators, Expressions			
CO2	Illustrates uses of conditional and iteration statements in python programs and understand the working of Strings in Python		
M2.01	To understand the use of conditional statements in Python	5	Understanding
M2.02	To understand the use of iteration statements in Python	5	Understanding

M2.03	To understand the use of Strings in Python	4	Understanding
	Series Test –I	1	
Contents: Control Statements- Selection structure -if, if-else, if-elif-else, Iteration with for/while loop, Basic coding skills –assignment, and comments Strings-Accessing characters, substrings, String methods			
CO3	Develop programs utilizing modules lists, dictionaries, sets, tuples.		
M3.01	To understand list, dictionaries, sets, tuples	8	Understanding
M3.02	To develop simple python programs to use list, dictionaries, sets, tuples	8	Applying
Contents: Lists - Basic list Operations and functions- Defining lists, accessing values in list, deleting values in list, updating lists, List of lists, Slicing, Searching and sorting Dictionaries - Dictionary functions, dictionary literals, adding and removing keys, accessing and replacing values, traversing dictionaries, Work with tuples. Sets			
CO4	Illustrate the use of numpy module and how to handle exceptions in python		
M 4.01	To understand the concept of numpy module in python	7	Understanding
M 4.02	To understand the concepts of exception handling	7	Understanding
	Series Test – II	1	
Contents: Numpy- Basics, Creating arrays, Arithmetic, Slicing, Random numbers Exceptions - Handle a single exception, handle multiple exceptions.			

Text / Reference

T/R	Book Title/Author
T1	Kenneth A Lambert., Fundamentals of Python : First Programs, 2/e, Cengage Publishing, 2016
T2	Harsh Bhasin, Problem Solving and Programming: First Edition, 2022
R1	Zed A Shaw, Learn Python 3 The Hard Way, Addison-Wesley, 2017
R2	Allen B. Downey, Think Python: How to Think Like a Computer Scientist, 2/e, Shroff, 2016
R3	Michael Urban and Joel Murach, Python Programming, Shroff/Murach, 2016
R4	Charles Severance. Python for Informatics: Exploring Information

Online Resources

Sl.No	Website Link
1	https://www.w3schools.com/python/